# Logarithmic Functions & Properties

## Section 1: Definition of Logarithms and Inverses

**1** Evaluate $\log_4\left(\frac{1}{64}\right)$.

**a.** $-3$

**b.** $3$

**c.** $-\frac{1}{3}$

**d.** $16$

**2** What is the domain of $f(x) = \ln(5 - 2x)$?

**a.** $(-\infty, 2.5)$

**b.** $(2.5, \infty)$

**c.** $(-\infty, 5)$

**d.** $(0, \infty)$

**3** Find the inverse function of $g(x) = 3 + 2\ln(x - 1)$.

**a.** $g^{-1}(x) = e^{(x-3)/2} + 1$

**b.** $g^{-1}(x) = e^{(x-3)/2} - 1$

**c.** $g^{-1}(x) = 2e^{x-3} + 1$

**d.** $g^{-1}(x) = e^{2(x-3)} + 1$

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## Section 2: Logarithmic Properties & Condensing Expressions

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**4** Express $2\log x - 3\log y + \frac{1}{2}\log z$ as a single logarithm.

**a.** $\log\left(\frac{x^2\sqrt{z}}{y^3}\right)$

**b.** $\log\left(\frac{2x\sqrt{z}}{3y}\right)$

**c.** $\log\left(x^2 - y^3 + \sqrt{z}\right)$

**d.** $\log\left(\frac{x^2}{y^3\sqrt{z}}\right)$

**5** If $\log_b 2 = 0.35$ and $\log_b 3 = 0.55$, what is the value of $\log_b 18$?

**a.** $1.45$

**b.** $0.90$

**c.** $1.25$

**d.** $1.10$

**6** Using the change of base formula, express $\log_5 12$ in terms of natural logarithms.

**a.** $\frac{\ln 12}{\ln 5}$

**b.** $\frac{\ln 5}{\ln 12}$

**c.** $\ln 12 - \ln 5$

**d.** $\ln\left(\frac{12}{5}\right)$

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## Section 3: Solving Exponential and Logarithmic Equations

**7** Solve for $x$: $3^{2x-1} = 27$.

**a.** $2$

**b.** $1$

**c.** $3$

**d.** $\frac{3}{2}$

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**8** Solve for $x$: $\log_2(x) + \log_2(x-2) = 3$.

**a.** $4$

**b.** $4$ and $-2$

**c.** $2$

**d.** $8$

**9** Solve for $t$: $50e^{0.04t} = 200$.

**a.** $\frac{\ln 4}{0.04} \approx 34.66$

**b.** $\frac{\ln 150}{0.04}$

**c.** $\frac{4}{0.04} = 100$

**d.** $\ln(4 \times 0.04)$

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## Section 4: Logarithmic Scales and Applications

**10** The pH of a chemical solution is given by $\text{pH} = -\log[\text{H}^+]$. If the hydrogen ion concentration is $1.0 \times 10^{-5}$ M, what is the pH?

**a.** $5.0$

**b.** $-5.0$

**c.** $1.0$

**d.** $9.0$

**11** An earthquake measures $7.2$ on the Richter scale, while a second measures $5.2$. How many times more intense is the first earthquake?

**a.** $100$

**b.** $2$

**c.** $20$

**d.** $1000$

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**12** What is the vertical asymptote of the graph of $f(x) = \log_3 (x + 4) - 2$?

**a.** $x = -4$

**b.** $x = 4$

**c.** $y = -2$

**d.** $x = 2$

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